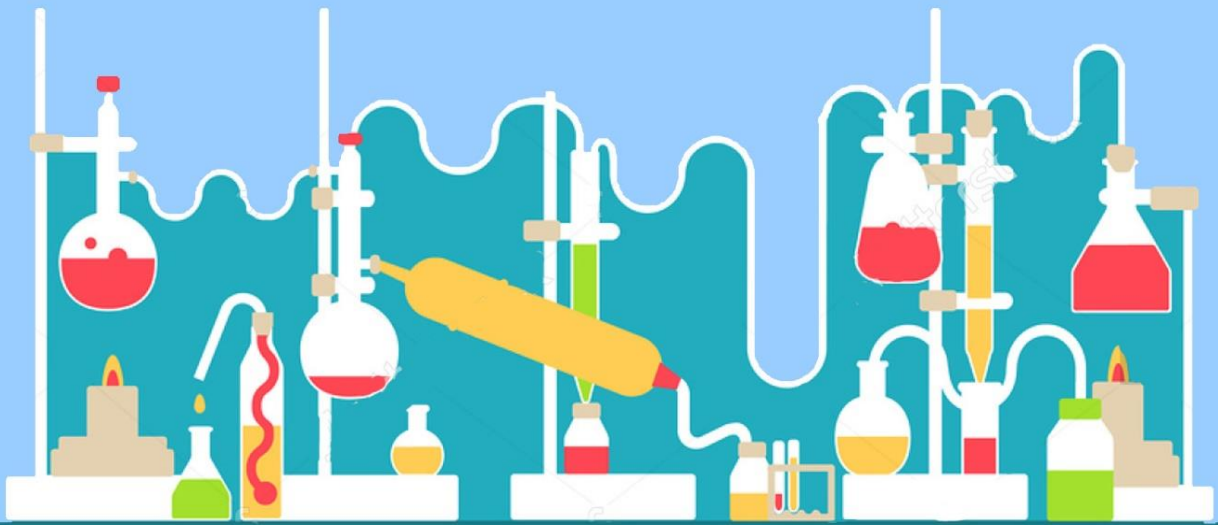




BIOLOGY LABORATORY



#4 Enzymes Activity



Experiment (4): Enzyme's Activity

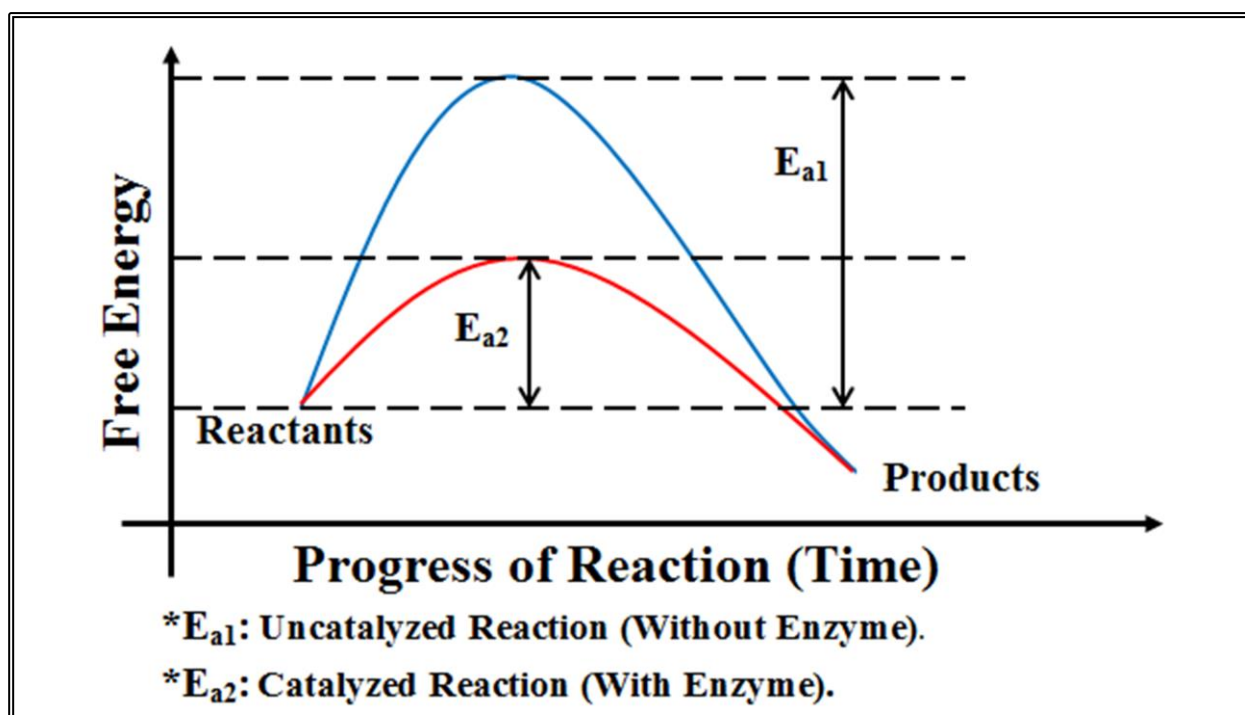
A) An Introduction to Enzymes

An *enzyme* (إنزيم) is defined as: A protein molecule that acts as a biological catalyst speeding up the rate of a reaction inside the cell, without being consumed.

- **How do enzymes work? How do they increase the rate of biological reactions?**

- Firstly, each reaction has an *activation energy* E_a (طاقة التنشيط) which is necessary to start breaking the chemical bonds within the reactants and form new chemical bonds of the products.

- Simply, enzymes lower (تقلل) that activation energy so the reaction goes faster. The figure below demonstrates the E_a of uncatalyzed and catalyzed reactions:



- **Few Notes Regarding Enzymes Structure & Function:**

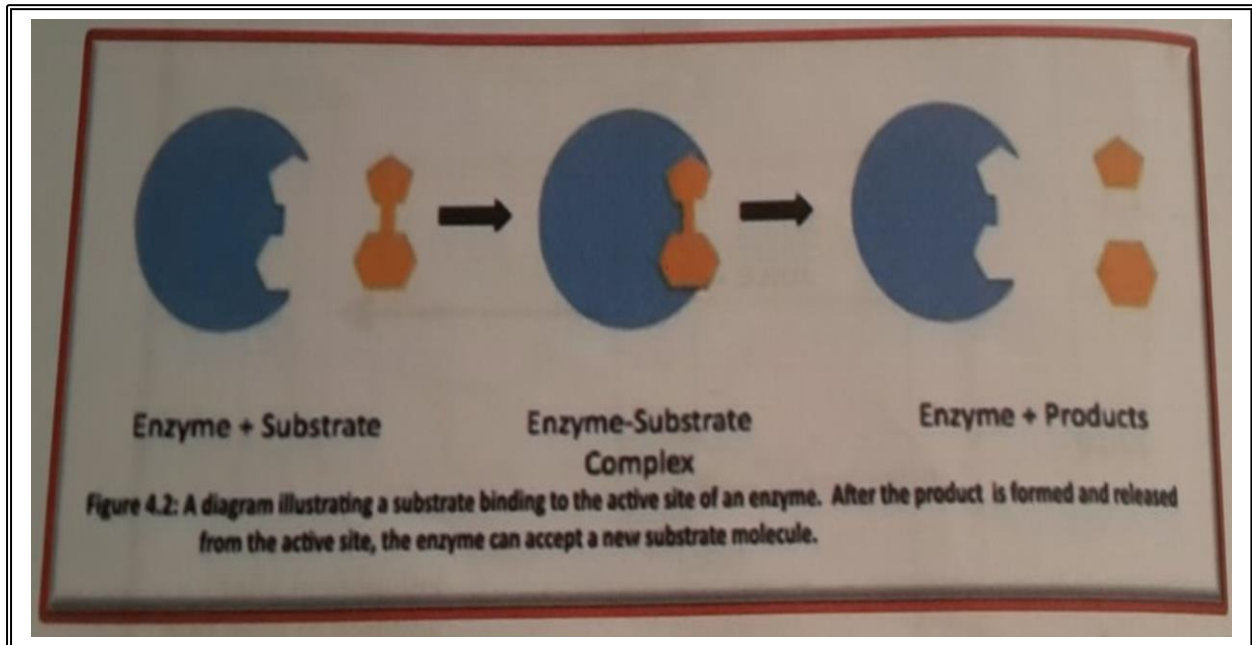
- Every enzyme is a protein (The opposite is incorrect). This means they have a 3-dimensional conformation.

- Because of this, each enzyme has an *Active Site* (المركز الفعّال); the portion of the enzyme where the *substrate* (المادة المتفاعلة) binds.

- The substrate binds to the enzyme at its active site forming an *enzyme-substrate complex*, then the substrate is converted into product(s).

- Enzymes are substrate specific; each enzyme acts upon a specific substrate and catalyzes its reaction. It typically will not work with any other substrate.

This figure shows enzyme structure and enzyme-substrate binding within an enzyme-catalyzed reaction:



- Enzymes can be used repeatedly; as we said in the definition, they are NOT consumed during the reaction.
- Enzyme's activity or function can be affected by several factors. We will discuss those in the next point.
- Enzyme's names usually end with the suffix (-ase), e.g., catalase, protease, amylase, etc.

B) Factors That Affect Enzyme's Activity (Reaction Rate)

1) Substrate Concentration

As the concentration of the substrate increases, the reaction rate will increase ONLY up to a certain limit; within the availability of the enzyme.

كلما زاد تركيز المادة المتفاعلة زادت سرعة التفاعل الذي ينشطه الإنزيم (لكن بشرط وجود كمية كافية من ذلك الإنزيم). إذا لم تكن كمية الإنزيم كافية ستظل سرعة التفاعل ثابتة مهما زاد تركيز المادة المتفاعلة.

2) Enzyme Concentration

As the concentration of the enzyme increases, the reaction rate will increase ONLY up to a certain limit; within the availability of the substrate...

كلما زادت كمية الإنزيم زادت سرعة التفاعل الذي ينشطه الإنزيم (لكن بشرط وجود تركيز كافٍ من المادة المتفاعلة). إذا لم يكن تركيز المادة المتفاعلة كافياً ستظل سرعة التفاعل ثابتة مهما زادت كمية الإنزيم.

3) Temperature

- Temperature can be defined as: the measure of the speed at which molecules are moving. Hence, as it increases, the speed of the molecule's movement will increase.

- Therefore, as the temperature increases the reaction rate increases, since substrate molecules are going to attach to their active site more often.

- However, enzyme-catalyzed reaction rates will increase also as the temperature increases but ONLY up to a certain limit. WHY?

→ Each enzyme has an *optimum* temperature which indicates the temperature degree at which the enzyme-catalyzed reaction undergoes at its highest rate.

→ Moving away from that optimum temperature (up or down), enzymatic activity decreases (The enzyme may be denatured if the temperature is far away from its optimum one), thus the enzyme-catalyzed reaction rate will decrease.

4) pH

- Similarly to the temperature, each enzyme has an *optimum* pH level at which it has the highest rate. When moving away from that optimum level, the enzymatic activity decreases (It also may be denatured if the change is huge).

- There is no proportion here.

C) Catalase Enzyme; an Example to Be Reviewed

Isolated From (Its Sources)	- Liver (الكبد); found within <i>peroxisomes</i> . - Potatoes; it is the most common source for laboratory purposes.
Acts Upon (The Substrate)	- Hydrogen Peroxide (H ₂ O ₂).
Optimum pH	- Approximately 7.0
Optimum Temperature	- Approximately 37°C
The Enzyme-Catalyzed Reaction	$2\text{H}_2\text{O}_2 + \text{Catalase} \rightarrow 2\text{H}_2\text{O} + \text{O}_2$ (Foam)

Multiple-choice Self-Quiz questions can be found in this link: <https://testmoz.com/900173>
Enter Your Name and Test Your Knowledge!

D) Test Your Understanding; Multiple-Choice Self-Quiz

Q-1) The portion of enzyme-substrate complex that is not used up during enzyme-catalyzed reactions is:

- a) the substrate.
- b) the enzyme.
- c) the active site.
- d) None of the above.

Q-2) Enzymes are _____ that function as biological catalysts by _____ the activation energy of metabolic reactions.

- a) carbohydrates, increasing.
- b) proteins, increasing.
- c) carbohydrates, lowering.
- d) proteins, lowering.

Q-3) Proteases break down:

- a) glycosidic linkages.
- b) peptide bonds.
- c) hydrogen bonds.
- d) ester bonds.

Q-4) Which of the following regarding enzyme-catalyzed reaction rate is correct?

- a) It increases as pH level increases generally.
- b) It increases as temperature increases then, it decreases.
- c) It decreases as substrate concentration increases up to certain limits.
- d) It is independent upon either sufficient amounts of enzyme or substrate.

Q-5) Which of the following regarding catalase enzyme is incorrect?

- a) It works under highly acidic conditions thus, it detoxifies alcohol and other poisons.
- b) It plays a crucial role in the liver; by minimizing hydrogen peroxide accumulation.
- c) The rate of the reaction catalyzed by itself would increase if pH was at about 7.
- d) It is commonly found within the liver and kidneys in high amounts.

KEY ANSWERS:

MCQ #	1	2	3	4	5
Answer	b	d	b	b	a

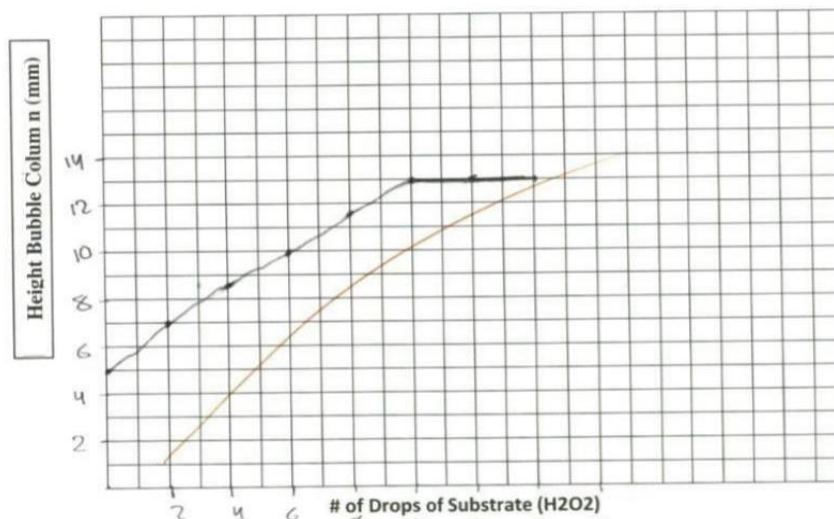
LABORATORY REPORT KEY ANSWERS

LABORATORY REPORT Enzymes

Table 4.1 Reaction Rates for Catalase at Various Substrate Concentrations

Test Tube No.	Potato extract (# drops)	H ₂ O ₂ (# drops)	Height of the foam Column (mm)
1	15	0.0	5
2	15	2	7
3	15	4	8.5
4	15	6	10
5	15	8	11.5
6	15	10	13
7	15	12	13
8	15	14	13

Fig. 4.1 Reaction Rate of Catalase as a Function of Substrate Concentration



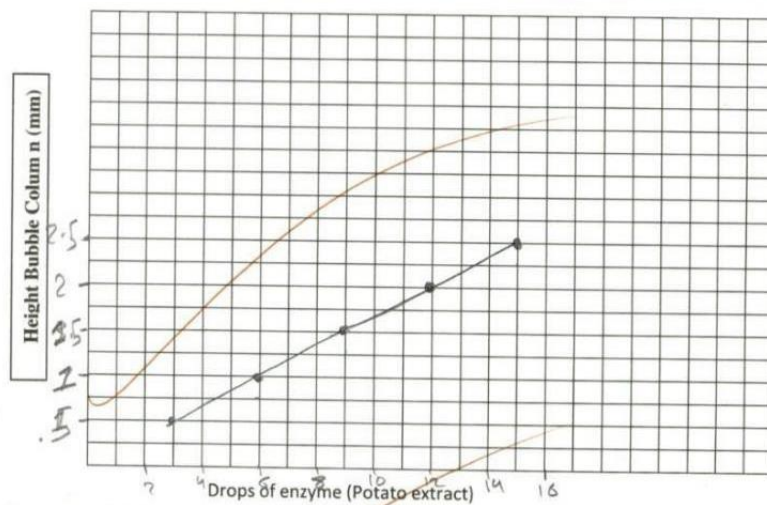
Make a general statement regarding the effect of substrate concentration on enzymatic reaction rate:

when we increase the substrate concentration the reaction rate will increase until it reaches the limit point where the concentration stops changing.

Table 4.2 Reaction Rate for Catalase at Various Enzyme Concentrations

Test Tube	H ₂ O ₂ (# drops)	Potato extract (#drops)	Height of the foam Column (mm)
1	20	0.0	0 ml
2	20	3	0.5 ml
3	20	6	1.0 ml
4	20	9	1.5 ml
5	20	12	2.0 ml
6	20	15	2.5 ml

Fig. 4.2 Reaction Rate of Catalase as a Function of Enzyme Concentration



Make a general statement regarding the effect of enzyme concentration on enzymatic reaction rate:

when we added and started increasing the enzymes concentration the reactions speed started to increase as well.

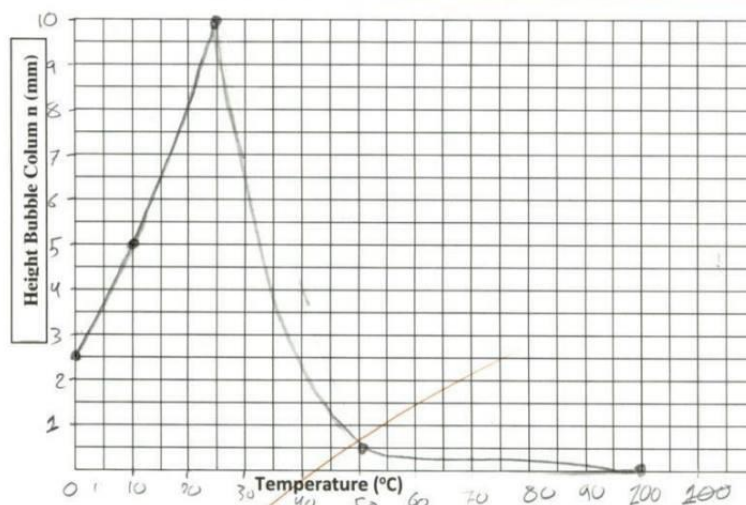


Table 4.3 Reaction Rates for Catalase at Various Temperatures

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Condition	Test Tube	H ₂ O ₂ (# drops)	Test Tube	Potato extract (# drops)	Height of the foam after pouring the test tubes together
Ice pocket	1	20	6	5	2.5 ml
Refrigerator	2	20	7	5	5 ml
Room Temperature	3	20	8	5	10 ml
Water bath (50 °C)	4	20	9	5	0.5 ml
Boiling bath	5	20	10	5	0 ml

Fig. 4.3 Reaction Rate of Catalase as a Function of Temperature



Make a general statement regarding the effect of temperature on enzymatic reaction rate:

As temperature increases reaction rate increases as

well until the optimal temperature (37°) after this the reaction rate will start to decrease.



Fig. 4.4 Reaction Rate of Catalase as a Function of pH V

Page 4 of Enzymes Report

Test Tube	10 drops of	Potato extract (# drops)	After 10 minutes	H ₂ O ₂ (#drops)	Height of the foam Column (mm)
1	Lemon juice	15		5	0 ml
2	H ₂ O	15		5	2 ml
3	NaOH	15		5	0 ml

Make a general statement regarding the effect of pH on enzymatic reaction rate:

the highest reaction rate was at optimum pH
(natural pH) which is 7
 as pH moves away from optimum the reaction rate will decrease.

GOOD LUCK



LABORATORY QUIZ KEY ANSWERS

Jordan University of Science and Technology

Department of Applied Biological Sciences

Biology Lab (107) Quiz

15

Q1: Choose the correct answer:

- 1- Which of the following factors affect enzyme activity:
A) pH B) temperature C) substrate concentration D) All are true
- 2- At high temperature the enzyme will be:
A) Stable B) denature C) inactive D) Give high activity
- 3- Complete the equation below which represent the H_2O_2 decomposition:
 $2H_2O_2 + \text{catalase} \rightarrow 2H_2O + O_2$
- 4- To indicate the activity of catalase we measure the height of one or more of the following:
A) CO_2 B) O_2 C) Foam D) B & C
- 5- At high and low pH the enzyme will be:
A) Active B) inactive C) inhibit D) denature

Q2: Which of the followings are True (T) or False (F)?

- 1- Catalase enzyme is found only in plant cells (F).
- 2- An enzyme's optimum temperature is that point which has the greatest reaction rate but does not denature the enzyme (T).
- 3- Enzymes catalyze more than one reaction (F).
- 4- As temperature increases, reaction rate will decrease and then increase (F).
- 5- As enzyme concentration increases the reaction rate increase and then inter the steady phase (F).

